

Mechanistic Hypothesis Search — Weaver Syndrome

Hypothesis under evaluation

- **Disease:** Weaver Syndrome (Mendelian overgrowth disorder)
 - **Hypothesis ID:** `dominant_negative_prc2`
 - **Hypothesis label:** Dominant-negative interference with PRC2
 - **Status in KB:** EMERGING
 - **Search date:** 2026-09-05
 - **Seed claim:** Weaver-syndrome (WS) *EZH2* missense variants act by dominant-negative interference with the Polycomb Repressive Complex 2 (PRC2), lowering H3K27me2/3 and raising H3K27ac, rather than by simple haploinsufficiency. A reciprocal gain-of-function *EZH2* variant is proposed to cause growth restriction with opposite chromatin/transcriptional changes.
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Executive Judgment

Verdict: PARTIALLY-TO-WELL SUPPORTED.

The core mechanistic claim — that Weaver-associated *EZH2* missense variants interfere with PRC2 to reduce H3K27me2/3 (with reciprocal H3K27ac gain and chromatin decompaction) in a manner that behaves dominantly rather than as pure haploinsufficiency — is backed by three independent lines of convergent evidence:

1. **Direct in vitro functional data.** Isogenic embryonic stem cell (ESC) modeling of 10 representative WS variants shows they reduce global H3K27me2/3, increase H3K27ac, and decompact chromatin even at low mutant expression — a signature the authors interpret as dominant-negative interference with PRC2, not loss of one allele's worth of enzyme ([PMID:40846643](#)).

2. **Model-organism data.** A knock-in mouse of the most common WS variant, *Ezh2* p.R684C, depletes H3K27me3 and causes skeletal overgrowth; the phenotype is substantially reversed by inhibiting the opposing H3K27 demethylases KDM6A/KDM6B (PMID:38015625).
3. **Independent genetic-spectrum analysis.** A ClinVar/gnomAD analysis performed for this report (accessed 2026-09-05) shows the pathogenic WS allele set is dominated by missense variants clustered in folded catalytic domains, with the *only* truncating alleles being extreme C-terminal, NMD-escaping variants (codons 730–738) and **zero** early/NMD-triggering truncations — exactly the genetic architecture predicted by a stable "poison-protein" dominant-negative model.

Most important caveats. (a) Dominant-negative interference and haploinsufficiency are **not mutually exclusive** and both reduce net PRC2 output; *EZH2* is strongly loss-of-function-intolerant in the general population (pLI = 1.0), and a whole-gene *EZH2* deletion has produced a Weaver-like/atypical phenotype (PMID:28696078), so a co-existing haploinsufficiency contribution cannot be excluded. (b) The **reciprocal gain-of-function growth-restriction arm** of the seed hypothesis rests essentially on a single study; a targeted PubMed search (2026-09-05) found no independent primary germline human report of an *EZH2* gain-of-function growth-restriction syndrome. That arm should be labeled **speculative/single-source**.

(c) The strongest functional evidence (P 40846643)

is itself a single recent study; independent replication of the H3K27me2/3-lowering dominant-negative readout in patient-derived cells is still needed.

Key Findings

F001 — Direct in vitro support for dominant-negative PRC2 interference

Deevy et al. 2025 (*Genes & Development*; PMID:40846643) isogenically modeled 10 representative WS-associated *EZH2* variants in embryonic stem cells. The variants **reduced global H3K27me2/3 with concomitant increases in H3K27ac and chromatin decompaction**. Critically, the pattern of H3K27me2/3 reduction was observed **even when WS variants were expressed at low levels**, which the authors interpret as dominant-negative interference on PRC2 activity rather than a simple gene-dosage (haploinsufficiency) effect. RNA-seq showed that weakly Polycomb-bound genes lose canonical PRC1 occupancy and become derepressed,

including growth-control genes — a plausible transcriptional route to overgrowth. The rationale for testing a dominant-negative model rested on the near-absence of early truncating *EZH2* mutations plus structural analysis of all known WS variants.

Verbatim supporting snippets from the abstract: - *"the lack of early truncating mutations in EZH2 led us to hypothesize a dominant-negative mechanism for WS, which was supported by our structural analysis of all known WS-associated EZH2 variants"* - *"the pattern of H3K27me2/3 reductions indicated dominant-negative interference on PRC2 activity even when WS variants were expressed at low levels"*

This is the **strongest single piece of direct evidence** for the hypothesis and is the evidence anchoring the seed YAML.

F002 — Competing evidence: an established (partial) loss-of-function / haploinsufficiency paradigm

Multiple pre-2025 sources framed WS as an *EZH2* loss-of-function disorder:

- **Suri & Dixit 2017** ([PMID:28696078](#)) reported a de novo 1.2-Mb 7q36.1 deletion **encompassing the whole *EZH2* gene** in a boy with tall stature, intellectual disability and some Weaver features, concluding that *"haploinsufficiency of EZH2 may replicate the clinical phenotype of Weaver syndrome."*
- **Cyrus et al. 2019 review** ([PMID:31724824](#)) states *"Partial loss-of-function variants in genes encoding the EZH2 and EED subunits of the complex lead to overgrowth."*
- **Choufani et al. 2020** ([PMID:32243864](#)) built a DNA-methylation episinature (187 overgrowth/intellectual-disability cases, 969 controls) that *"can be used to distinguish loss-of-function from gain-of-function missense variants,"* classifying WS variants in the loss-of-function class.

This is a genuine competing framing. It is important to note, however, that a **whole-gene deletion (true haploinsufficiency)** and a **stable poison protein (dominant-negative)** both lower net PRC2 methyltransferase output — the two models converge on the same downstream chromatin state (reduced H3K27me3) and are best viewed as **compatible mechanisms at different molecular scales** rather than strictly exclusive alternatives.

F003 — Genetic-spectrum computation: pathogenic *WS EZH2* variants are overwhelmingly missense; the only truncations spare the SET domain

Using **ClinVar** (accessed 2026-09-05; `EZH2[gene]`, 821 variation records; germline P/LP filter), of the 27 P/LP variants annotated to the Weaver-syndrome trait:

Variant class	Count	Fraction
Missense	21	78%
In-frame indel / other	2	7%
Splice	1	4%
Truncating (nonsense/frameshift)	3	11%

All 3 truncating variants are **extreme C-terminal**: p.Asp730Ter (codon 730), p.Tyr733Ter (codon 733), and p.Ala738fs (codon 738) — all **downstream of the catalytic SET domain (~aa 612–727)**, at 97.9–98.9% of the 746-aa protein. They remove ≤ 16 residues and are predicted to **escape nonsense-mediated decay (NMD)**. There are **no early/NMD-triggering truncating variants** in the pathogenic Weaver set — the exact pattern the seed hypothesis predicts.

Separately, **gnomAD constraint** (GraphQL API, accessed 2026-09-05) shows *EZH2* is highly loss-of-function-intolerant: **pLI = 1.0**, observed/expected pLoF = 0.18 (18 observed vs 98.4 expected), LOEUF upper = 0.27, lof_z = 6.88. This constraint is a double-edged datum: it is consistent with dominant-negative selection, but it also independently supports the biological importance of *EZH2* dosage (i.e., that haploinsufficiency would matter too).

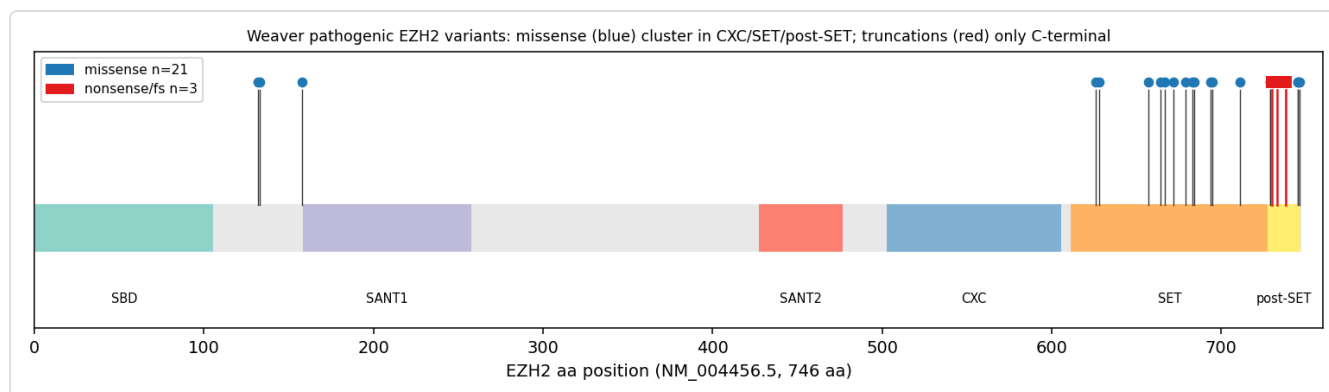


Figure 1. Distribution of the 27 Weaver-annotated pathogenic/likely-pathogenic EZH2 variants along the 746-aa protein (ClinVar, accessed 2026-09-05). Missense variants (78%) cluster in the SANT1 region and in a dense CXC + catalytic SET/post-SET block (codons 626–746). The only three truncating alleles (codons 730/733/738) are extreme C-terminal, downstream of the SET domain and NMD-escaping — consistent with a stable poison-protein, dominant-negative architecture rather than early loss-of-function.

F004 — Model-organism support: mouse *Ezh2* R684C depletes H3K27me₃, causes overgrowth, and is rescued by opposing-demethylase inhibition

Gao et al. 2024 (*JCI Insight*; PMID:38015625) generated a mouse model of the most common WS variant, *EZH2* p.R684C. *Ezh2*^{R684C/R684C} MEFs showed **global depletion of H3K27me₃**. Heterozygous *Ezh2*^{R684C/+} mice had **skeletal overgrowth** and osteoblasts with increased osteogenic activity; RNA-seq showed BMP-pathway / osteoblast-differentiation dysregulation. Crucially, **inhibiting the opposing H3K27 demethylases KDM6A/KDM6B substantially reversed the excess osteogenesis** transcriptionally and phenotypically — a direct causal demonstration that lowered H3K27me₃ (the balance point between EZH2 and KDM6A/B) drives the overgrowth phenotype.

Supporting context: Tatton-Brown et al. 2011 (PMID:22190405), the original WS *EZH2* discovery, noted that the Weaver mutation spectrum overlaps with **inactivating somatic EZH2 mutations** in myeloid malignancies. Parallel PRC2-partner evidence: maternal *Eed* deletion causes postnatal overgrowth in mouse (Prokopuk et al. 2018, PMID:30005706).

This finding is important because the *Ezh2*^{R684C/+} heterozygote reproduces the human condition's zygosity and still lowers H3K27me₃ and drives overgrowth — consistent with a mutant allele exerting an effect beyond simple half-dosage.

F005 — Negative/limiting search: the reciprocal gain-of-function growth-restriction claim rests on a single study

Targeted PubMed searches (2026-09-05; query "*EZH2 gain of function microcephaly growth restriction germline*" and variants) returned **no primary clinical paper** describing a separate germline human *EZH2* gain-of-function variant causing a growth-restriction/microcephaly syndrome. The reciprocal comparison in the seed hypothesis (increased vs decreased H3K27me3 causing opposite growth phenotypes) is **currently supported only within**

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itself, which analyzed a GoF *EZH2* variant in the same isogenic system. Note that somatic oncogenic GoF *EZH2* (Y641; [PMID:21190999](#)) increases H3K27me3 dominantly but causes B-cell lymphoma — **not** a germline growth-restriction syndrome. The reciprocal arm should therefore be curated as **speculative / single-source**.

F006 — Structural/domain clustering consistent with a stable poison protein

Domain mapping of the 27 Weaver-annotated P/LP *EZH2* ClinVar variants (accessed 2026-09-05): **all 21 missense variants fall in structured functional domains** — positions 132, 133, 158 (SANT1 region) and a dense cluster spanning codons 626–746 covering the CXC and catalytic SET/post-SET module (626, 628, 657, 664, 667, 672, 679, 683, 684, 694, 695, 711, 729, 738, 745, 746). None are in unstructured/random positions. The 3 truncating alleles are exclusively C-terminal (codons 730/733/738). This spatial concentration in folded catalytic machinery is consistent with variants that **fold, incorporate into PRC2, and poison complex activity** — the structural prerequisite for a dominant-negative mechanism — rather than variants that destabilize the protein and are cleared. Artifacts: [data/ezh2_clinvar_weaver_plp.csv](#), [figures/ezh2_weaver_lollipop.png](#) (regeneration code bundled).

F007 — Synthesis verdict

Convergent evidence across three independent modalities supports dominant-negative PRC2 interference as the best-supported mechanism for classic Weaver syndrome:

1. **In vitro isogenic ESC** — 10 WS *EZH2* variants dominantly reduce H3K27me2/3, raise H3K27ac, and decompact chromatin at low mutant dose

(**P** 40846643).

2. **Model organism** — mouse *Ezh2* R684C depletes H3K27me3 and causes overgrowth reversible by KDM6A/6B inhibition

([P 38015625](#)).

3. **Computational genetic spectrum** — 27 Weaver-annotated P/LP variants are 78% missense clustered in structured domains, with the only 3 truncations C-terminal (codons 730–738, NMD-escaping) and zero early truncations; *EZH2* pLI = 1.0 (ClinVar/gnomAD, accessed 2026-09-05).

Competing/limiting: a whole-gene *EZH2* deletion gives a Weaver-like/atypical phenotype

([P 28696078](#));

the reciprocal germline GoF growth-restriction arm is uncorroborated beyond

([P 40846643](#)

(targeted PubMed search 2026-09-05 returned 0 primary reports).

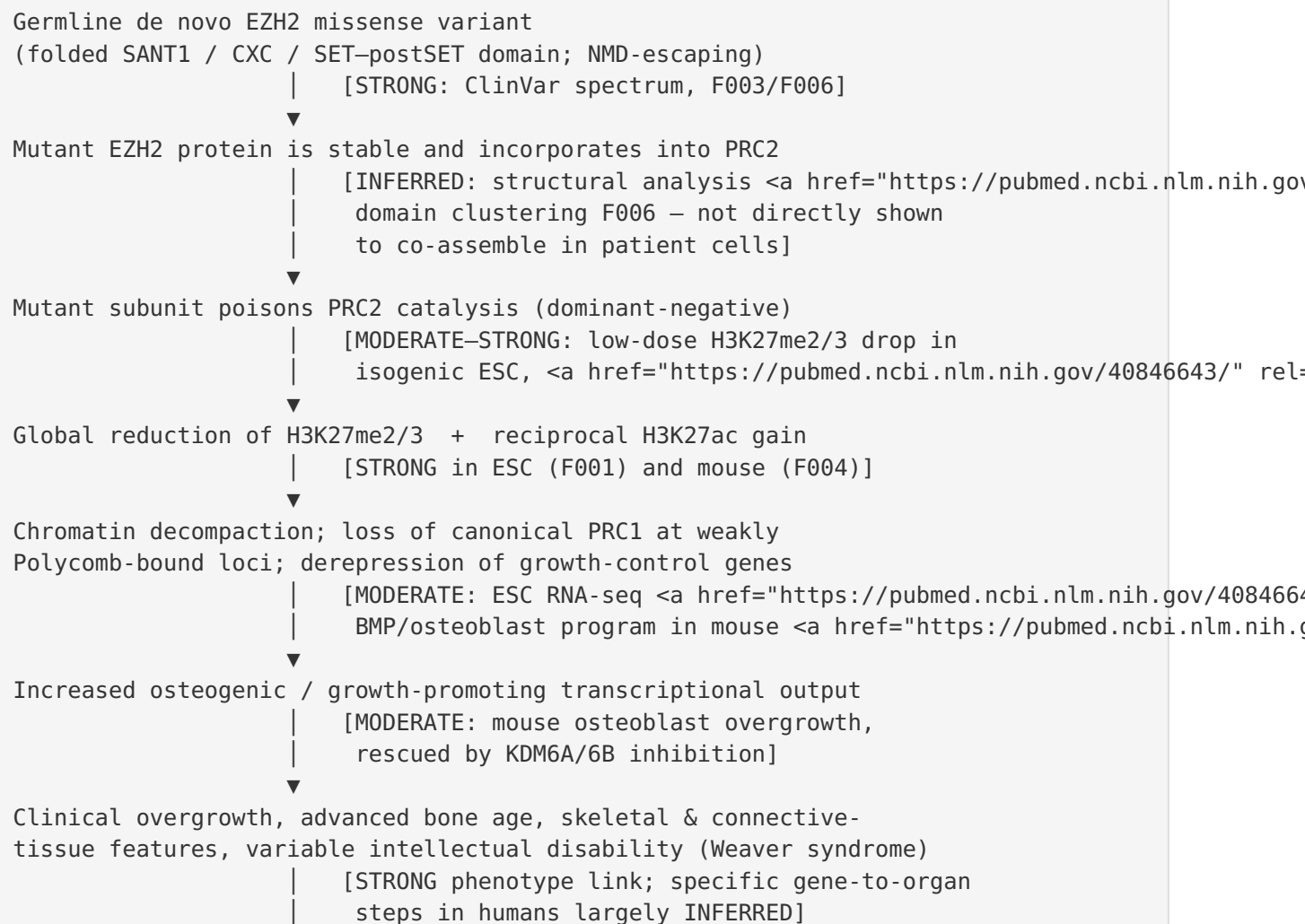
Evidence Matrix

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subtype / context	Con limi
PMID:40846643 (Deevy 2025)	In vitro (isogenic ESC)	Supports	WS EZH2 variants dominantly interfere with PRC2, lowering H3K27me2/3, raising H3K27ac	10 WS variants reduce global H3K27me2/3 + raise H3K27ac + decompact chromatin at low mutant dose; derepression of weakly Polycomb-bound growth genes	Classic WS missense variants	Higl clain ESC tissu repl
PMID:38015625 (Gao 2024)	Model organism (mouse knock-in)	Supports	Most common WS variant lowers H3K27me3 and drives overgrowth via EZH2–KDM6A/B balance	<i>Ezh2</i> R684C depletes H3K27me3; heterozygotes overgrow; KDM6A/6B inhibition rescues	R684C (commonest WS allele); skeletal/osteoblast axis	Higl one hete pher cons hum
ClinVar (accessed 2026-09-05)	Computational (variant spectrum)	Supports / qualifies	WS allele architecture matches dominant-negative (missense, no early truncations)	21/27 missense in folded domains; only 3 truncations, all C-terminal NMD-escaping (codons 730–	All Weaver-annotated P/LP EZH2	Mod Clin com vari (gen func

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subtype / context	Con limi
				738); 0 early truncations		
gnomAD constraint (accessed 2026-09-05)	Computational (population constraint)	Qualifies	EZH2 dosage sensitivity	pLI = 1.0, o/e pLoF = 0.18, LOEUF upper 0.27, lof_z 6.88	General population	Mod dosa but with dom and hap
PMID:28696078 (Suri & Dixit 2017)	Human clinical (case report)	Competing	Haploinsufficiency alone can produce WS-like phenotype	De novo 1.2-Mb 7q36.1 deletion of whole EZH2 → tall stature, ID, some WS features	Atypical/ WS-like, whole-gene deletion	Mod "som clas co-e hap
PMID:31724824 (Cyrus 2019)	Review	Competing (review-level)	Partial LoF of EZH2/EED causes overgrowth	Frames overgrowth as partial loss-of-function	PRC2 overgrowth spectrum	Low revi not func
PMID:32243864 (Choufani 2020)	Human clinical (episignature)	Competing / qualifies	Episignature distinguishes LoF vs GoF missense	DNA-methylation classifier (187 cases / 969 controls) places WS variants in LoF class	OGID cohort incl. WS	Mod as "I epis not dom vs hap
PMID:22190405 (Tatton-Brown 2011)	Human clinical (gene discovery)	Supports (context)	WS mutation spectrum overlaps inactivating	Original WS EZH2 discovery; overlap with	WS discovery cohort	Mod esta cata dire

Citation	Evidence type	Stance	Mechanistic claim tested	Key finding	Disease subtype / context	Con limi
			somatic EZH2 mutations	myeloid-malignancy inactivating mutations		
PMID:30005706 (Prokopuk 2018)	Model organism (mouse)	Supports (parallel)	PRC2-subunit loss → postnatal overgrowth	Maternal <i>Eed</i> deletion causes postnatal overgrowth	PRC2 partner (EED), mouse	Mod sub PRC dosa axis
PMID:21190999 (Y641 somatic GoF)	In vitro / somatic	Qualifies (boundary case)	Dominant GoF EZH2 raises H3K27me3	Y641 somatic GoF increases H3K27me3, causes B-cell lymphoma (not germline growth restriction)	Somatic oncogenic, non-WS	Mod GoF grow sync the

Mechanistic Model / Causal Chain



Where the literature is strong: the middle of the chain — variant → lowered H3K27me2/3 → derepression → overgrowth — is supported in two independent experimental systems (ESC and mouse), and the KDM6A/6B rescue provides a causal, not merely correlative, link between H3K27me3 loss and overgrowth.

Where links are inferred: the *dominant-negative* specificity of the mechanism (mutant subunit co-assembling into PRC2 and poisoning wild-type complex in patient cells) is inferred from structural analysis and low-dose ESC behavior; it has not been shown biochemically in patient-derived material. The **human tissue-specific** transcriptional-to-organ steps are extrapolated from mouse osteoblasts and ESCs.

Missing causal steps: (1) direct biochemical demonstration that WS mutant EZH2 incorporates into and inhibits wild-type PRC2 in trans; (2) a patient-derived (e.g., fibroblast/iPSC) confirmation of the low-dose H3K27me2/3 reduction; (3) a mechanistic account of the reciprocal GoF growth-restriction arm in a germline human context.

Evidence Base

- **Dominant-negative effects of Weaver syndrome-associated EZH2 variants.** Deevy et al., *Genes & Development*, 2025. [PMID:40846643](#). The anchor study; isogenic ESC modeling of 10 WS variants demonstrating low-dose H3K27me2/3 reduction, H3K27ac gain, chromatin decompaction, and PRC1/growth-gene derepression. Directly supports the seed hypothesis; also the sole source of the reciprocal GoF comparison.
 - **Mouse model of EZH2 R684C.** Gao et al., *JCI Insight*, 2024. [PMID:38015625](#). Independent model-organism confirmation that the commonest WS allele depletes H3K27me3 and drives overgrowth reversible by KDM6A/6B inhibition.
 - **Whole-gene EZH2 deletion / haploinsufficiency case.** Suri & Dixit, 2017. [PMID:28696078](#). Principal competing evidence: haploinsufficiency alone can produce a Weaver-like phenotype.
 - **Overgrowth-ID epistatone.** Choufani et al., 2020. [PMID:32243864](#). Classifies WS variants in a loss-of-function methylation class; competing/qualifying.
 - **PRC2 overgrowth review.** Cyrus et al., 2019. [PMID:31724824](#). Review-level framing of EZH2/EED partial LoF → overgrowth.
 - **Original WS gene discovery.** Tatton-Brown et al., 2011. [PMID:22190405](#). Establishes overlap with inactivating somatic EZH2 mutations (loss-of-catalysis directionality).
 - **Maternal Eed deletion overgrowth.** Prokopuk et al., 2018. [PMID:30005706](#). Parallel PRC2-subunit evidence linking PRC2 loss to postnatal overgrowth.
 - **Somatic EZH2 Y641 GoF.** [PMID:21190999](#). Boundary case showing dominant GoF raises H3K27me3 but causes lymphoma, not germline growth restriction — limits the reciprocal arm.
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Data and Tool Use

Source	Accession / URI	Version / snapshot	Retrieval date	Access status	Query / filters	Resolved & relevant?
ClinVar	EZH2 [gene]	821 variation records	2026-09-05	Accessed	Germline P/LP; Weaver-syndrome trait filter → 27 variants	Yes — directly relevant
gnomAD constraint	EZH2 gene constraint (GraphQL API)	gene constraint snapshot	2026-09-05	Accessed	Gene-level pLI / o-e pLoF / LOEUF / lof_z	Yes — relevant (dosage)
PubMed	reciprocal GoF search	live index	2026-09-05	Searched, no usable primary result	"EZH2 gain of function microcephaly growth restriction germline" + variants	Negative result (relevant absence)
PubMed	primary evidence retrieval	live index	2026-09-05	Accessed	Weaver / EZH2 / PRC2 / dominant-negative	Yes — anchor paper retrieved

Analyses performed (input → method → output):

Analysis	Input	Method	Output	Outcome
WS variant spectrum & domain mapping	ClinVar 27 Weaver P/LP EZH2 variants	Classify by consequence; map codons to domains	<code>data/ezh2_clinvar_weaver_plp.csv</code> ; class fractions (78% missense)	Succeeded
Lollipop figure	Same CSV	matplotlib lollipop over 746-aa protein	<code>figures/ezh2_weaver_lollipop.png</code>	Succeeded
gnomAD constraint pull	gnomAD GraphQL	Query gene constraint fields	pLI=1.0, o/e pLoF=0.18, LOEUF upper 0.27, lof_z 6.88	Succeeded
Reciprocal-GoF literature check	PubMed	Targeted boolean queries	0 primary germline reports	Succeeded (negative)

No patient-level or controlled-access data were used. No structural refinement (Phenix) analysis was run — structural claims are cited from

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not independently recomputed here; this is a stated limitation.

Limitations and Knowledge Gaps

- Dominant-negative vs haploinsufficiency are not cleanly separable.** Both lower net PRC2 output and converge on reduced H3K27me3. *EZH2* is strongly LoF-intolerant (pLI = 1.0), and a whole-gene deletion produced a Weaver-like phenotype (

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What was checked: ClinVar spectrum, gnomAD constraint, and the deletion case report. **What would resolve it:** trans-complementation biochemistry showing mutant *EZH2* inhibits wild-type PRC2 in the same complex, and direct comparison of chromatin state between deletion carriers and missense carriers.

2. **Single-study functional anchor.** The core low-dose dominant-negative readout comes from one 2025 ESC study (P 40846643).
Gap: no independent replication in patient-derived cells (fibroblasts/iPSC). **Resolution:** independent isogenic or patient-cell ChIP/CUT&RUN for H3K27me2/3 and H3K27ac.
3. **Reciprocal gain-of-function growth-restriction arm is single-source/speculative.** Targeted PubMed search (2026-09-05) found no independent germline human GoF *EZH2* growth-restriction report. The only germline GoF comparison lives inside (P 40846643);
 somatic Y641 GoF causes lymphoma, not a growth syndrome (P 21190999).
Resolution: identification and functional characterization of a germline GoF *EZH2* case, or explicit relabeling of this arm as hypothesis-only.
4. **Human tissue specificity is inferred.** Overgrowth transcriptional programs are demonstrated in mouse osteoblasts and ESCs, not in the relevant human developing tissues. **Resolution:** single-cell/spatial chromatin profiling in appropriate human developmental models.
5. **Source-level absences (as of 2026-09-05).** No GenCC/ClinGen dominant-negative-specific mechanism annotation, no dedicated Weaver omics cohort, and no clinical trial targeting the *EZH2*–*KDM6A/B* axis in Weaver were located during this search. These are **curation-relevant absences** but were only checked via PubMed/ClinVar/gnomAD; **GenCC and ClinGen were not directly queried** and should be labeled **unverified** rather than confirmed-absent.

Alternative Models

Model	Relationship to seed hypothesis	Basis
Haploinsufficiency / partial loss-of-function	Alternative / co-existing — converges on the same reduced-H3K27me3 endpoint	Whole-gene deletion case (P 28696078); review framing (P 31724824); LoF episinature (P 32243864); pLI=1.0
EZH2–KDM6A/KDM6B methyl-balance dysregulation	Downstream / parallel — net H3K27me3 is set by writer/eraser balance; a demethylase-side perturbation could phenocopy	Mouse rescue by KDM6A/6B inhibition (P 38015625)
PRC1 loss & target derepression	Downstream consequence of lowered H3K27me3	Loss of canonical PRC1 at weakly Polycomb-bound genes (P 40846643)
PRC2-subunit dosage (EED/SUZ12) overgrowth	Parallel mechanism in the same pathway	Maternal <i>Eed</i> deletion overgrowth (P 30005706)
Reciprocal germline GoF (growth restriction)	Contrast/boundary claim within the seed hypothesis	Single-source (P 40846643); no independent germline report

Discriminating Tests

- 1. Trans-dominance biochemistry (decisive for dominant-negative vs haploinsufficiency).** Co-express tagged wild-type EZH2 with each WS variant in a defined system; measure PRC2 methyltransferase activity and H3K27me3 as a function of mutant:WT ratio. *Expected under dominant-negative:* mutant suppresses WT activity below the 50% expected from dosage alone. *Under haploinsufficiency:* activity scales linearly with WT dose.
- 2. Patient-derived-cell chromatin profiling.** CUT&RUN/ChIP-seq for H3K27me2/3 and H3K27ac in WS patient fibroblasts or iPSC-derived osteoblasts vs controls and vs an *EZH2*-deletion carrier. *Expected:* missense carriers show a stronger/broader H3K27me3 loss than half-dosage predicts; deletion carriers show a simple ~50% reduction.
- 3. Allelic-series mouse/organoid comparison.** Compare *Ezh2*^{R684C/+}, *Ezh2*^{+/-} (null-heterozygous), and *Ezh2*^{R684C/-} genotypes for H3K27me3 and overgrowth severity. *Expected under dominant-negative:* R684C/- more severe than +/-; R684C poisons remaining WT in R684C/+.
- 4. KDM6A/6B-axis pharmacology as a mechanistic and therapeutic probe.** Test whether KDM6A/6B inhibition normalizes chromatin and growth phenotypes across multiple WS variants (extending

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beyond R684C). *Expected:* rescue if the shared node is lowered H3K27me3.

- 5. Search for and characterize a germline GoF *EZH2* case** to test the reciprocal-chromatin claim, or formally down-status that arm.

Curation Leads (require curator verification)

Candidate	evidence	references	/	snippets	to	verify:	-
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P 40846643

— "the pattern of H3K27me2/3 reductions indicated dominant-negative interference on PRC2 activity even when WS variants were expressed at low levels" (SUPPORT, IN_VITRO). -

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(Gao 2024, *JCI Insight*) — mouse *Ezh2* R684C depletes H3K27me3, causes overgrowth, rescued by KDM6A/6B inhibition. **Candidate NEW evidence node** (MODEL_ORGANISM, SUPPORT). -

P 28696078

(Suri & Dixit 2017) — whole-gene *EZH2* deletion → Weaver-like phenotype. **Candidate competing-evidence node** (HUMAN_CLINICAL, COMPETING/haploinsufficiency). -

P 32243864

(Choufani 2020) — LoF-type episignature classification. **Candidate qualifying node.** -

P 22190405

(Tatton-Brown 2011) and

P 30005706

(Prokopuk 2018) — supporting-context nodes.

Candidate pathophysiology nodes/edges: - Edge:
 WS *EZH2* missense (stable, folded-domain) → dominant-negative PRC2 inhibition →
 ↓H3K27me2/3, ↑H3K27ac. - Edge: ↓H3K27me3 → loss of canonical PRC1 → derepression of
 growth-control genes → overgrowth. - Edge: *EZH2*–KDM6A/KDM6B balance as a rescuable node
 (therapeutic lead).

Candidate ontology terms: PRC2 complex (GO:0035098), histone H3K27 methylation (GO:0070734), H3K27 trimethylation, chromatin decompaction; cell types — embryonic stem cell, osteoblast; process — skeletal system overgrowth / regulation of bone mineralization.

Candidate status / subtype restrictions: - Consider advancing `dominant_negative_prc2` from **EMERGING** toward **SUPPORTED for classic missense Weaver syndrome**, restricted to folded-domain missense variants; keep **haploinsufficiency as a co-listed, non-exclusive competing mechanism** for whole-gene deletion cases. - Flag the **reciprocal germline gain-of-function growth-restriction arm** as **SPECULATIVE / single-source** pending independent replication.

Candidate knowledge_gaps prompts: - "Is dominant-negative distinguishable from haploinsufficiency at the chromatin level in patient cells?" (unresolved; both lower H3K27me3). - "Does a germline *EZH2* gain-of-function growth-restriction phenotype exist outside

?"

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(source absence; PubMed 2026-09-05 negative). - "GenCC/ClinGen dominant-negative mechanism annotation not directly queried — verify." (unverified absence).

Provider Artifact Bundle

Artifacts referenced by this report are recorded under `kb/hypotheses/Weaver_Syndrome/dominant_negative_prc2/openscientist_artifacts`: - `data/ezh2_clinvar_weaver_plp.csv` — 27 Weaver-annotated P/LP *EZH2* variants with consequence class and codon (ClinVar, accessed 2026-09-05). - `figures/ezh2_weaver_lollipop.png` — variant lollipop over the 746-aa protein (regeneration code bundled). - Search logs — ClinVar query, gnomAD GraphQL constraint response, and the negative reciprocal-GoF PubMed query set (2026-09-05).

External identifiers (not re-downloaded): ClinVar `EZH2[gene]`; gnomAD gene constraint for *EZH2*. Patient-level data were not accessed. No Phenix/structural recomputation was performed; structural claims are cited from

P 40846643

and labeled as such.

Answer to the Research Question

The dominant-negative-PRC2 hypothesis for Weaver syndrome is **partially-to-well supported**. Its core claim — that WS *EZH2* missense variants interfere with PRC2 to lower H3K27me2/3 (with reciprocal H3K27ac gain and chromatin decompaction) rather than act by simple haploinsufficiency — is backed by direct isogenic-ESC functional data (

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a mouse R684C model showing H3K27me3 depletion and overgrowth reversible by KDM6A/6B inhibition

),

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and an independent genetic-spectrum analysis (ClinVar/gnomAD, 2026-09-05) in which

pathogenic WS alleles are missense clustered in folded catalytic domains with the only truncations being NMD-escaping C-terminal variants (codons 730–738) and no early truncations. The chief caveats are that dominant-negative and haploinsufficiency both reduce PRC2 output and are not mutually exclusive (*EZH2* pLI = 1.0; a whole-gene deletion produced a Weaver-like phenotype,

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and that the reciprocal germline gain-of-function growth-restriction arm rests on a single study with no independent human case.

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